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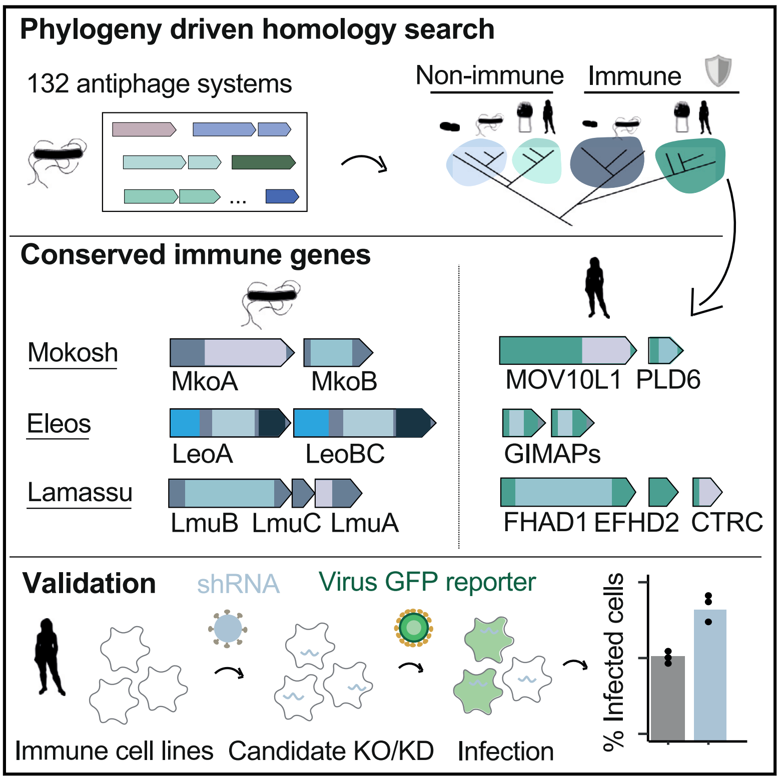
Conservation of antiviral systems across domains of life reveals immune genes in humans

Graphical abstract



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In brief

A subset of eukaryotic immune proteins originates from antiphage systems protecting bacteria from viruses. Cury et al. demonstrate that homology search of antiphage systems in eukaryotic genomes can be utilized to discover human immune proteins. This work highlights how evolutionary genomics can illuminate eukaryotic immunology.

Highlights

- A subset of eukaryotic immune proteins originates from prokaryotic antiphage systems
- Phylogeny-driven search identifies eukaryotic immune proteins linked to bacteria
- Anti-transposon piRNA pathway proteins originate from antiphage Mokosh
- GIMAPs and FHAD1/CTRC are human antiviral effectors from antiphage origin

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Conservation of antiviral systems across domains of life reveals immune genes in humans

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SUMMARY

Deciphering the immune organization of eukaryotes is important for human health and for understanding ecosystems. The recent discovery of antiphage systems revealed that various eukaryotic immune proteins originate from prokaryotic antiphage systems. However, whether bacterial antiphage proteins can illuminate immune organization in eukaryotes remains unexplored. Here, we use a phylogeny-driven approach to uncover eukaryotic immune proteins by searching for homologs of bacterial antiphage systems. We demonstrate that proteins displaying sequence similarity with recently discovered antiphage systems are widespread in eukaryotes and maintain a role in human immunity. Two eukaryotic proteins of the anti-transposon piRNA pathway are evolutionarily linked to the antiphage system Mokosh. Additionally, human GTPases of immunity-associated proteins (GIMAPs) as well as two genes encoded in microsynteny, FHAD1 and CTRC, are respectively related to the Eleos and Lamassu prokaryotic systems and exhibit antiviral activity. Our work illustrates how comparative genomics of immune mechanisms can uncover defense genes in eukaryotes.

INTRODUCTION

Viral infections have profound implications for human health, agriculture, and ecosystems, necessitating in-depth investigations into the immune mechanisms that govern eukaryotic virus control. Components of eukaryotic immunity are regularly discovered, indicating that our understanding of eukaryotic defense systems remains incomplete. Technological innovation often drives the discovery of immune proteins, as illustrated by the development of genome-wide CRISPR screens that contributed significantly to the fine dissection of human antiviral immunity.[1] Most eukaryotes of agricultural or ecological importance are not suitable for such approaches due to the lack of model organisms and the limitations of experimental tools. Developing innovative methodologies to identify eukaryotic mechanisms of defense would not only shed light on unexplored angles of human immunology but also enable to decipher immunity in non-model organisms.

Recently, dozens of bacterial systems have been shown to play immunological roles against phages and other mobile genetic elements.[2–7] Each system may be composed of unique or multiple genes whose products act in concert. Genes of a given system are closely located on the genome, allowing tight coregulation of gene expression and genetic exchange via horizontal gene transfer.[6,7] Uncovering antiphage systems led to the discovery that a subset of these proteins is likely at the origin of immune proteins of eukaryotes, including humans.[8–17] Antiviral immunity in humans relies on steady-state expression of restriction factors as well as on the early detection of pathogens by specific receptors.[18] For instance, infection by DNA viruses, such as herpesviruses, is detected by cyclic GMP-AMP synthase (cGAS), which signals through stimulator of interferon (IFN) genes (STING).[19,20] The cGAS/STING pathway of eukaryotes likely originates from the bacterial CBASS system, which is involved in antiphage immunity through the production of cyclic nucleotides.[10,15,21] cGAS activation translates into the production of inflammatory cytokines (e.g., IFNs), which upregulate the expression of IFN-stimulated genes (ISGs), a broad panel of antiviral effectors that include the viperin protein.[18,22,23] Human and other eukaryotic viperins originate from the prokaryotic

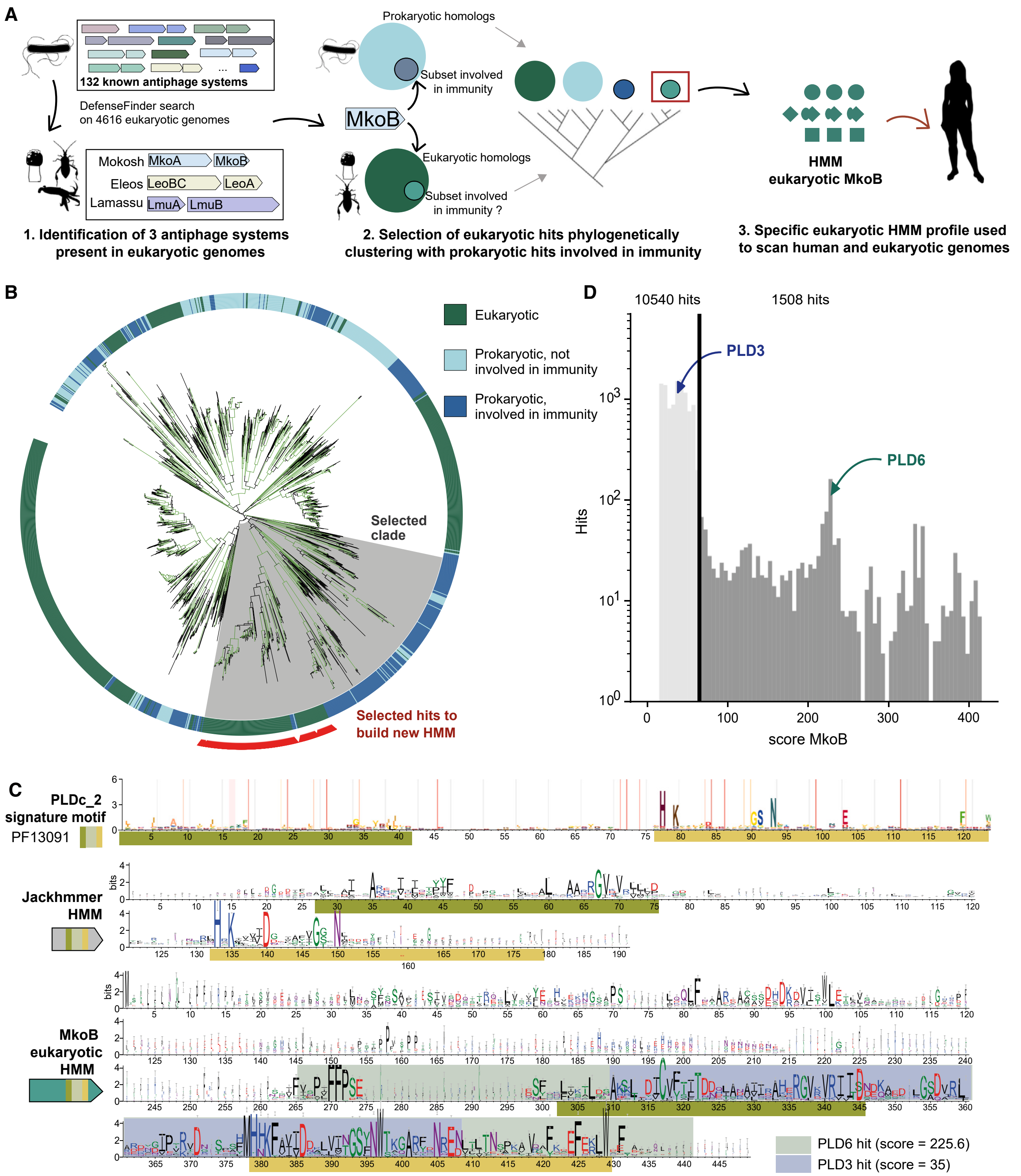

Figure 1. A phylogeny-driven homology search to identify potential eukaryotic immune proteins

(A) A sequence homology and phylogeny-based approach identifies eukaryotic proteins with sequence similarity to antiphage systems.

(B) Phylogenetic tree obtained by analysis of prokaryotic and eukaryotic databases using a prokaryotic MkoB HMM profile (step 2 of the approach). Ultra-fast bootstrap >95 are indicated in green. The clade highlighted in gray is selected for further analysis because it regroups prokaryotic antiphage proteins and eukaryotic proteins with similarity that cluster together. Selected hits used to build a eukaryotic MkoB HMM profile are indicated in red.

(legend continued on next page)

viperins, with which they share a common antiviral mechanism.[9] Additional examples of sequence similarity between antiphage and eukaryotic immune proteins have been recently unraveled, including prokaryote/eukaryote likely homologs of gasdermins and nucleotide-binding oligomerization domain, leucine rich repeat-containing proteins (NLRs), which play a key role in eukaryotic cell death upon infection.[11–14,24] In these examples, it is the pre-existing knowledge of eukaryotic immune proteins that enabled the identification of sequence and/or structure similarities with newly discovered antiphage systems.

In this work, we leverage a reverse approach to discover eukaryotic proteins of defense, including antiviral proteins, by sequence similarity with bacterial antiphage systems. Focusing on humans as a case study, we couple homology search to phylogeny to retrieve previously unknown evolutionary relationships between prokaryotic and eukaryotic systems of defense. Our data document the conservation of immunity across domains of life and illustrate how it can be harnessed for the discovery of immune genes in eukaryotes.

RESULTS

Phylogeny-driven search of eukaryotic homologs of antiphage systems

To discover eukaryotic proteins playing a role in immunity, we hypothesized that (1) proteins involved in newly discovered antiphage systems may be conserved in eukaryotes and that (2) these eukaryotic proteins with sequence similarity to antiphage systems may have retained immune functions. We thus established a method to discover eukaryotic proteins with sequence similarity to bacterial antiphage proteins. Previous work demonstrated that such similarity can be detected using amino acid sequence conservation. This is the case for viperins, whose sequences are highly conserved between bacteria and eukaryotes, including humans.[9] Bacterial and human gasdermins, on the other hand, show little sequence similarity but are conserved at the structural level.[13] Sequence similarity carried by proteins can thus be lost across extended evolutionary periods. Sequence conservation, however, exists between bacterial and fungal gasdermins, illustrating that likely homologs of antiphage systems could be detected by analyzing a breadth of diverse eukaryotic genomes.

We recently developed DefenseFinder, a bioinformatics pipeline designed to identify known antiphage systems in bacterial genomic datasets.[2] We repurposed it to uncover proteins with sequence similarity to these systems in eukaryotic genomes. The pipeline detects 132 types of antiphage systems by means of protein sequence homology. Each component of a system is represented by a protein profile, allowing for low sequence similarity detection (hidden Markov model [HMM]). We applied DefenseFinder to a database of 4,616 genomes spanning the kingdom of eukaryotes and detected 14,702 protein hits, which we subsequently filtered to remove bacterial genomic contaminations (see STAR Methods). These hits correspond to eukaryotic proteins that share enough homology with proteins of antiphage systems to be detected by DefenseFinder. Most hits correspond to antiviral genes with known prokaryote-eukaryote conservations, such as viperins (904 hits) and Argonautes (10,248 hits), demonstrating that DefenseFinder can indeed uncover eukaryotic homologs of known antiphage systems. In addition to these positive controls, we detect putative homologs of the recently discovered antiphage systems Mokosh (178 hits), Eleos (415 hits), and Lamassu (624 hits).[3] Such conservation suggests that, akin to viperins or CBASS, antiphage Mokosh, Eleos, and Lamassu have eukaryotic counterparts (Figure 1A).

Our initial analysis is hindered by DefenseFinder being tailored to the analysis of prokaryotic genomes. Indeed, DefenseFinder includes criteria of genomic proximity to detect proteins similar to a given system. For example, Mokosh is composed of the proteins Mokosh A (MkoA) and MkoB. DefenseFinder will detect the presence of Mokosh in a given eukaryotic genome only if proteins with sequence similarity to MkoA and MkoB are both detected and encoded by genes next to each other on the genome. To bypass this issue, we performed homology searches of the individual components of Mokosh, Lamassu, and Eleos using their respective HMM profiles.

Antiphage proteins are composed of domains that are not necessarily specific to immune pathways but can be shared by other proteins unrelated to immunity.[25] For example, MkoB (Figure 1A), part of antiphage Mokosh, bears a phospholipase D (PLD) domain. This domain is also present in non-immune proteins such as ClsB, which is involved in lipid synthesis. Consequently, performing homology search for single antiphage proteins will likely retrieve a mix of immune and non-immune eukaryotic proteins. We hypothesized that prokaryotic and eukaryotic proteins similar in sequence with immune functions would share a more recent common ancestor compared with non-immune proteins. In that framework, phylogeny could efficiently discriminate between potentially immune hits and other unrelated hits. We constructed a pipeline that pairs HMM search with a phylogenetic analysis (Figure 1A). First, prokaryotic and eukaryotic proteins similar in sequence to a given antiphage protein are retrieved using a sensitive HMM sequence search (bacterial protein profile). A phylogeny is then constructed, incorporating prokaryotic and eukaryotic sequences (Figures 1A and 1B). To focus specifically on eukaryotic proteins that are likely homologs of antiphage proteins, we isolate eukaryotic sequences that are phylogenetically close to bacterial antiphage sequences and construct an HMM profile from these proteins. Eukaryotic sequences phylogenetically close to non-immune prokaryotic proteins are discarded. The newly built HMM profile, theoretically enriched for eukaryotic antiphage homologs, is then used to query a eukaryotic database, retrieving a set of eukaryotic hits, including human proteins (Figure 1A).

We first benchmarked the ability of our method to identify proteins that play a role in immunity. When bacterial MkoB proteins

(C) Weblogos comparisons of (1) the PLDc_2 motif from PFAM, (2) the Jackhmmer HMM profile generated by analysis of bacterial MkoB (3 iterations, see STAR Methods), and (3) the eukaryotic MkoB HMM profile obtained with our pipeline. Green and yellow indicate two separate motifs of the PLD signature. Human PLD6 and PLD3 are mapped on the MkoB eukaryotic HMM.

(D) Score distribution of the custom eukaryotic MkoB profile against eukaryotic database. Chosen threshold is indicated by a black line.

See also Figure S1 and Tables S1, S2, and S4.

are used as an input, our approach retrieves a single human hit, PLD6, a mitochondrial cardiolipin hydrolase. PLD6 is the only identified candidate among 11 human proteins carrying a PLDc_2 domain detected by the protein profile PFAM13091 (PLD1–5, FAM83B-D-F-G, and PGS1, Figures 1C and 1D). PLD6 has been documented to participate in the piRNA pathway, which protects the mammalian germ line from the deleterious expression of transposable elements (TEs).[26] Our pipeline is thus able to identify human proteins with a documented immune role through sequence similarity to antiphage proteins. With respect to an analysis relying on bacterial HMM profiles, our query using refined eukaryotic profiles is expected to retrieve a more specific set of hits, as it putatively encloses an additional layer of phylogenetic information. It is also expected to be more sensitive, as the profile should capture eukaryotic sequence diversity. As such, only 17.4% of hits are common between the two methods (Figure S1).

Eukaryotic proteins with sequence similarities to antiphage defense systems display immune functions

To interrogate the specificities of our method, we compared it to jackhmmer and foldseek, two state-of-the-art methods of remote homology analysis based on sequence and structure, respectively.[27,28] When provided with the bacterial MkoB displaying the best HMM score from the DefenseFinder profile as an input, jackhmmer retrieves 9,870 eukaryotic hits, including 4 human hits (PLD1 to PLD4). A similar query of the human genome using foldseek retrieves PLD1 to PLD4 as the best hits, in a total of 16 hits. Our pipeline, based on phylogeny, thus identifies a specific set of hits compared with jackhmmer and foldseek, as illustrated by their respective weblogos (Figure 1C). Weblogo representation of the sequences forming the HMM profiles indeed indicates that while jackhmmer mostly captures the signature of the PLDc_2 domain, our pipeline incorporates more information that allows the identification of PLD6 (Figure 1C; STAR Methods).

We extend this observation by comparing jackhmmer's and our method's analyses of individual components of Mokosh (MkoA and MkoB), Eleos (LeoA and LeoBC), and Lamassu (LmuA and LmuB). Profiles obtained with our pipeline are longer than jackhmmer's profiles, allowing them to be more specific, which is instrumental in reducing the number of hits for experimental validation (Figure S2; STAR Methods). Human homologs retrieved with these analyses were then evaluated for documented roles in immunity (Figures 2A and S2; Tables S1 and S2). Jackhmmer identifies 821 hits, with 7% participating in immune defense (58 hits). By comparison, our method identifies a total of 168 human hits, 21% of which have established roles in immunity (35 hits, Figure 2A). This includes EHD4, MOV10, and GIMAP2, previously involved in IFN signaling.[29] There is thus a 3-fold increased detection of immune genes when using our pipeline compared with state-of-the-art jackhmmer.

In addition, we investigated the recent evolutionary history of the human proteins showing sequence similarity to antiphage systems isolated by our method and jackhmmer. To do so, we assessed their evolutionary patterns across mammals by measuring for each gene: (1) the degree of nucleotide sequence conservation (GerpRS) in coding exons, as a measure of selective constraint and (2) the average rate of non-synonymous to synonymous divergence across all branches of the mammalian phylogeny (branch-site random effects likelihood model [BS-REL]), as a measure of adaptive evolution. We compared genes identified by our pipeline or jackhmmer with a list of human defense genes (i.e., documented as antiviral, antibacterial, and antifungal genes by Gene Ontology) and all human genes as a negative control. We show that, similar to defense genes, genes identified with our method present both a lower conservation (GerpRS, Wilcoxon rank-sum two-sided $p = 5.6 \times 10^{-5}$) and a higher rate of adaptation than genome-wide genes (BS-REL, Wilcoxon rank-sum two-sided $p = 1.8 \times 10^{-32}$), which is not the case for genes isolated with jackhmmer (Figure 2B; STAR Methods).[30] To further strengthen our claim, we computed BS-REL scores across various branches of the mammalian tree (primates, glires, cetartiodactyla, and zooamata, Figure 2C). In contrast to genes identified by jackhmmer, the genes pinpointed by our method display high proportion of adaptive mutations across all branches considered, similar to what is observed for known defense response genes (if not higher). This supports the ability of our pipeline to identify defense genes with increased specificity.

In the following parts of the manuscript, we explore the immune role of the identified human hits (Table S1; STAR Methods).

Proteins protecting the animal germ line are similar to the Mokosh antiphage system

Mokosh is composed of proteins bearing at least two domains: an RNA helicase (carried by MkoA) and a PLD domain (present in MkoB, Figures 3A and 3B). In addition to uniquely identifying PLD6 as the human homolog of MkoB, our pipeline uncovers MOV10 and its paralog, MOV10L1, as homologs of MkoA (Figure 3A). MOV10L1 and PLD6 are well-documented proteins of the piRNA pathway, which prevents the expression of TEs in the animal germ line.[26] Tightly controlling TEs is instrumental in germ cells, as their expression can potentially disrupt genomic architecture and dysregulate gene expression.[31] Consequently, animals deficient for the piRNA pathway are sterile.[26,32] TE regulation relies on the generation of piRNAs, a type of small RNAs, by cleavage of piRNA precursors originating from genomic transcription (Figure 3C). MOV10L1 unwinds piRNA precursors via its RNA helicase domain to permit cleavage by the nuclease PLD6. This piRNA production activity depends on the physical interaction between the two proteins (Figure 3C).[26,33,34]

To explore the evolutionary relationship between Mokosh and the piRNA pathway, we built a phylogeny that includes (1) bacterial MkoB proteins bearing a PLDc_2 domain, (2) eukaryotic homologs detected by our method, and (3) an outgroup composed of eukaryotic and prokaryotic ClsB, which are non-immune proteins involved in lipid synthesis. Phylogeny of MkoB/PLD6 shows that the eukaryotic clade detected through our pipeline, including human PLD6, emerges from a subgroup of MkoB (in green in Figures 3B and S3A). The picture is less clear when considering the phylogeny of MkoA/MOV10L1, due to the high abundance of eukaryotic MkoA-like proteins, presumably derived from multiple events of gene duplication. Nonetheless, we observe that MkoA and MOV10L1 likely share a bacterial common ancestor, branching separately from an outgroup composed of eukaryotic and bacterial proteins (Figure S3A). The topology of such phylogenies is consistently reproduced

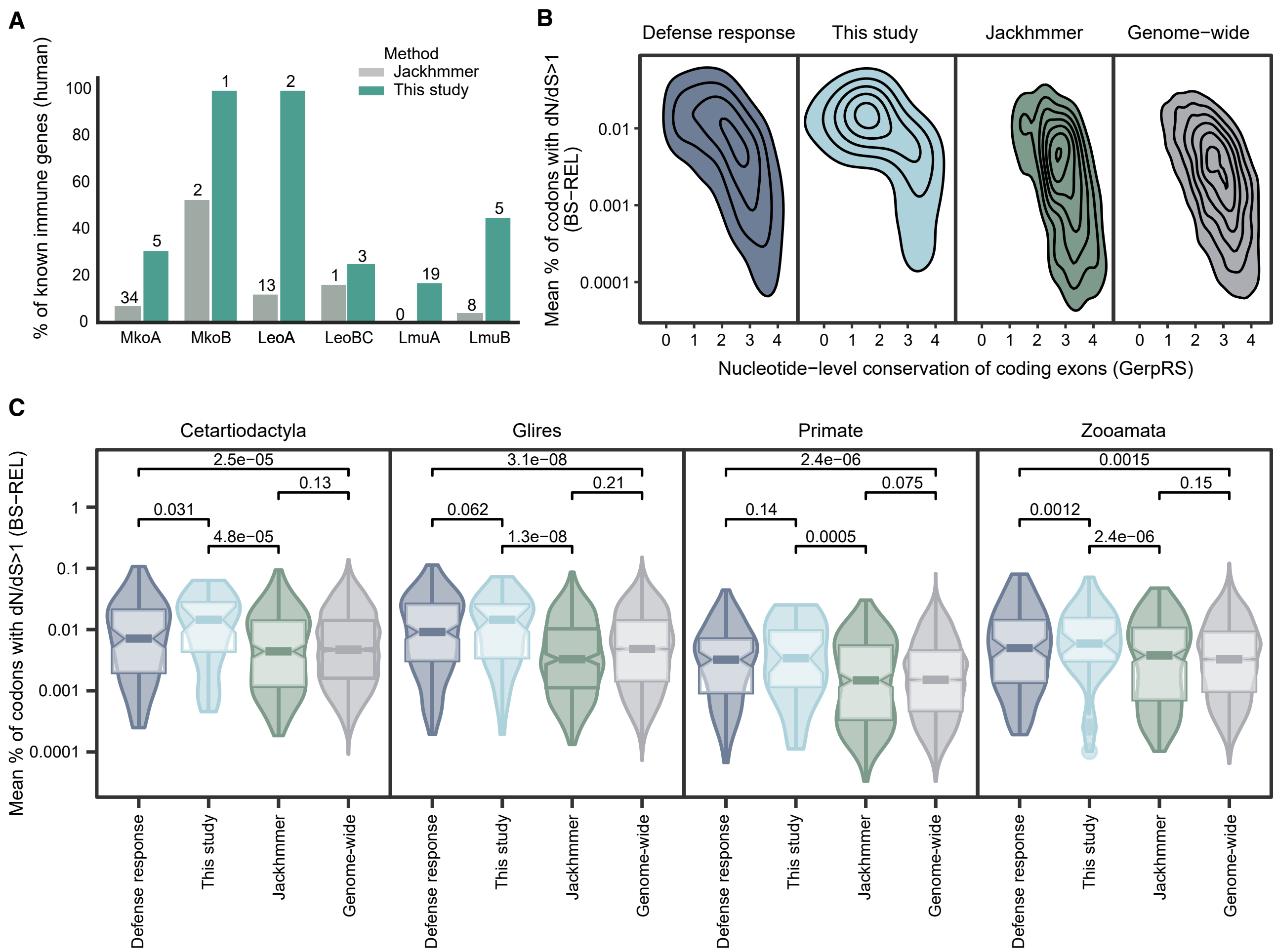

Figure 2. A phylogeny-driven homology search identifies eukaryotic proteins with immune functions

(A) Percentage of known immune genes in human hits obtained by jackhmmer and the method used in this study, as determined by manual curation. Absolute numbers of proteins with sequence similarity with a documented immune function are indicated.

(B) Recent evolution of human genes is measured by (1) the degree of nucleotide sequence conservation in coding exons (GerpRS), measuring selective constraint, and (2) the average rate of non-synonymous to synonymous divergence across all branches of mammalian phylogeny (BS-REL), measuring adaptive evolution. GerpRS and BS-REL values of human immune genes (positive control defined by Gene Ontology), total human genes (negative control), and genes isolated by our pipeline or jackhmmer are plotted. Genes identified with our pipeline show lower conservation (GerpRS) and a higher rate of adaptation (BS-REL), measured by Wilcoxon rank-sum test (two-sided p = 5.6e−5 and 1.8e−32, respectively). Scores can be found in Table S5.

(C) Compared rate of adaptive mutations between hits identified by jackhmmer or our pipeline in this study across 4 different branches of the mammalian tree. For each branch, we provide the rate of adaptive mutations genome-wide and among defense response genes as a reference.

See also Figure S2 and Tables S1, S2, and S5.

by different methods of multiple sequence alignment (see STAR Methods).

These results strongly suggest that a key step of the piRNA pathway—piRNA production—is evolutionary linked to the antiphage system Mokosh. We next evaluated if distant sequence homology between Mokosh and PLD6/MOV10L1 translates to structure conservation. Structural alignments of conserved domains of human PLD6 and MOV10L1 with a Mokosh system from *Escherichia coli* ETEC H10407 are shown in Figures 3D and S3B–S3F. Protein similarity can be quantified by template modeling (TM) scores. A TM score above 0.5 typically indicates that the predicted fold may be similar to the true structure.[36,37] A variant of this score, ipTM, evaluates the quality of the structure prediction at protein-protein interfaces (ipTM > 0.8 high-quality predictions, ipTM < 0.6 likely failed predictions).[38]

The PLD domain of PLD6 shows similarities to the bacterial Mokosh (TM score 0.63), as does the RNA helicase domain of MOV10L1, though to a lesser extent (TM score of 0.43, Figure 3D). While the RNA helicase and PLD domains are conserved in MOV10L1 and PLD6, respectively, the proteins also bear additional domains that are not present in bacterial Mokosh systems (Figures 3D and S3B). Expanding structural comparisons to other eukaryotes similarly show that domains of both proteins have clear structural homologies with prokaryotic MkoB and MkoA (Figures S3C–S3F; TM scores in Table S3). It was previously established that PLD6 and MOV10L1 physically interact in a complex that produces piRNAs.[26,33,34] As no experimental

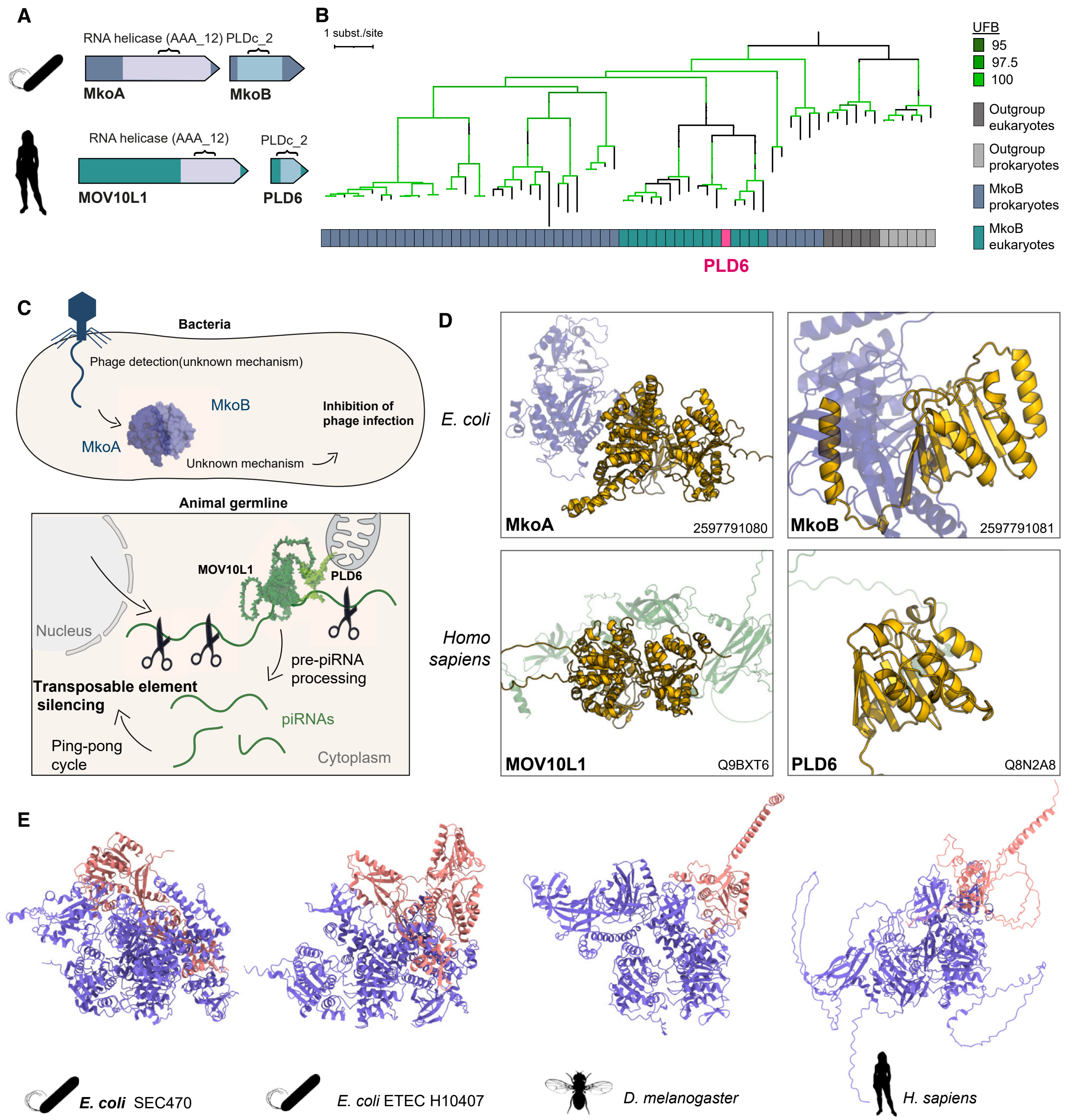

Figure 3. PLD6 and MOV10L1, two proteins of the piRNA pathway guarding the animal germ line, likely originate from Mokosh antiphage systems

(A) The bacterial antiphage system Mokosh is composed of the proteins MkoA and MkoB. MOV10L1 and PLD6 are likely human homologs of MkoA and B, detected by our method, which show conservation of the RNA helicase and PLD domains, respectively.

(B) Phylogeny of bacterial MkoB and their eukaryotic proteins with sequence similarity, including human PLD6 (pink). Ultra-fast bootstraps (UFBs) >95 are highlighted in green. Full tree in Figure S3A.

(C) In bacteria, the Mokosh system detects and thwarts phage infection through unknown mechanisms. In the animal germ line, repression of deleterious TE expression relies on the cleavage of precursor piRNAs (pre-piRNAs) by MOV10L1 and PLD6 to generate piRNAs that are used to sequence-specifically target TEs.

(D) Conservation of the RNA helicase and PLD-like nuclease domains of prokaryotic antiphage MkoA and MkoB (blue) in the human homologs MOV10L1 and PLD6 (green). IMG identifiers are indicated for bacterial Mokosh and AlphaFold DB IDs for MOV10L1 and PLD6. Optimal local alignment between the two

(legend continued on next page)

data are available for bacterial Mokosh proteins, we co-folded MkoA and MkoB of two experimentally validated Mokosh using Alphafold-Multimer (Figure 3E).[3] A physical interaction between MkoA and MkoB is supported by ipTM scores of 0.80 and 0.84 for Mokosh from strains SEC470 and HT10407, respectively (Figure 3E). To interrogate if such interaction is maintained across domains of life, we co-folded two pairs of eukaryotic MkoA/B (MOV10L1/PLD6 from *Drosophila melanogaster* and *Homo sapiens*) and showed that, while protein structures evolve, an interaction between MkoA and MkoB homologs seems to be conserved from prokaryotes to humans (Figure 3E).

Our data demonstrate that MOV10L1 and PLD6 are likely homologs of antiphage MkoA and B. Mokosh thus provided key genes for the evolution of the piRNA pathway, constituting another example of contribution of antiphage systems to this pathway, in addition to Argonautes, another family of proteins pivotal for the piRNA pathway originating from bacterial immunity.[39]

GIMAPs, human proteins with sequence similarity to antiphage Eleos, are antiviral

We next harnessed our pipeline to identify antiviral proteins in humans. Antiphage Eleos is a two gene-system (LeoA and LeoBC) coding for proteins bearing dynamin-like domains from the P-loop NTPase superfamily (Figures 4A and 4B).[3,40] Using LeoA as a seed, our pipeline retrieves two human hits, mitofusin-1 and -2, known to be involved in immunity (Table S1). Mitofusins also regulate mitochondria dynamics, illustrating that proteins part of defense mechanisms can perform pleiotropic roles outside of immunity.[41] When using LeoBC as an input, we detect 12 human hits, including EHD4, a known antiviral ISG, and GTPases of immunity-associated proteins (GIMAPs), a group of 8 paralogs that bear an MMR_HSR1 domain from the P-loop NTPase superfamily (among 98 human proteins carrying an MMR_HSR1 domain [PFAM01926.24]; Figures 4A and S4; Table S1).[29,42] GIMAPs play a role in immunity in mammals, as loss of GIMAP5 in rats translates in decreased levels of T lymphocytes, and expression of GIMAP4, 5, and 6 in rat macrophages impairs infection by the parasite *Toxoplasma gondii*.[43] AIG1 and AIG2, phylogenetically related to GIMAPs, play a role in antibacterial defense in plants.[44] Structural comparison between HY001, a LeoBC of *Bacillus sp.*, and GIMAP6 displays moderate structural similarities (TM score of 0.45, Figure 4C).

To evaluate the evolutionary relationship between Eleos and GIMAPs, we generated a phylogeny incorporating (1) bacterial LeoBC proteins, (2) a subset of eukaryotic proteins detected by our method, and (3) an outgroup composed of prokaryotic and eukaryotic GTPases termed *E. coli* Ras-like proteins (ERA, Figures 4B and S4A). This phylogeny shows that GIMAPs belong to a monophyletic clade of eukaryotic proteins that emerges from prokaryotic LeoBC, highlighting that GIMAPs likely originate from the Eleos antiphage system (Figure 4B).

In humans, GIMAPs are ISGs whose transcription is induced by type-I and type-II IFNs (Figure S5A).[45] We first interrogated the antiviral role of endogenous GIMAP5 and 6 by knockdown in the human epithelial cell line HeLa and infection with GFP-encoding herpes simplex virus 1 (HSV-1). The experiment was performed at steady state as well as in cells pre-treated with IFN to increase the levels of GIMAPs. Diminishing the levels of GIMAP6 in cells not treated with IFN translates into a 70% increase in the percentage of infected cells, a 50% increase in viral replication, and 11.5-fold increased viral particle production (Figures 5A–5C). The reduction of GIMAP5 levels had more subtle antiviral effects with a 2.5-fold change in viral titers (Figures 5B and 5C). Similar results were obtained in cells pre-treated with IFN, demonstrating that GIMAP5 and 6 are antiviral.

To further understand the antiviral role of GIMAPs, we used a system of transient expression via plasmid transfection in human embryonic kidney 293T cells (Figure 5A). After transfection, cells are infected with HSV-1. Levels of plasmid-driven protein expression, as well as infection, are monitored at the single-cell level by immunostaining and flow cytometry (Figure 5A). Expression of the positive control MxB, an antiviral factor blocking HSV-1, results in a 25% reduction in the percentage of infected cells and a 50% reduction in GFP mean fluorescence intensity (MFI), a proxy for viral protein accumulation (Figure 5D).[46] Expression of a negative control ISG15, another ISG with no expected antiviral effect in these conditions, is inefficient in blocking HSV-1 infection (Figure 5D).[47] Cells expressing GIMAP6 display a 50% reduction in infection by HSV-1, as well as a 50% reduction in the GFP intensity (Figures 5D and S5B–S5D). GIMAP5 shows variable antiviral activity depending on the experiments (Figures 5D and S5C). GIMAP5 and 6 do not display, in these experimental conditions, an antiviral effect against vesicular stomatitis virus, a negative-strand RNA virus (Figure S5E).

The activity of GIMAPs—and other P-loop NTPase proteins from the dynamin-like family—has been documented to rely on the formation of dimers, whose dissociation depends on GTP hydrolysis.[48,49] GIMAPs indeed carry a GTPase domain composed of G-box motifs (Figure S4B). Mutating a key lysine residue of the G-box 1 inhibits GTP hydrolysis, potentially trapping the dimers in an active state.[48,49] We thus hypothesized that preventing GTP hydrolysis in GIMAP5 and 6 would enhance antiviral activity. GIMAP5 and 6 carrying a mutated lysine in the G-box 1 motif (K40A and K53A, respectively) display 70% reduction in infection compared with control (Figures 5D, S4B, S5B, and S5D). Locking GIMAP5 and 6 in an active state thus enhances antiviral capacity compared with wild-type proteins.

Finally, as the GTPase domains of bacterial Eleos and human GIMAPs are both immune and evolutionary linked, we assessed if the bacterial GTPase could substitute for the human domain. We focused on GIMAP6, in which the strongest antiviral effect was observed, as well as on the experimentally validated Eleos system from *Bacillus vietnamensis* NBRC 101237.[3] In this Eleos system, the GTPase domain of LeoA is marginally more similar to GIMAP6 than LeoBC and encodes for well-annotated G1 to G5 boxes. We thus constructed a GIMAP-Eleos chimera

structures (as determined by foldseek) is represented in yellow. Part of the structures are omitted and shown in Figures S3B–S3F. Structures of the bacterial Mokosh were predicted using AlphaFold2.[35] PDB identifiers, TM, and ipTM scores can be found in Table S3.

(E) Structural predictions of the interaction between MkoA (blue) and MkoB (pink) homologs across domains of life, as determined by Alphafold Multimer. ipTM scores of 0.80 (*E. coli* ETEC H10407), 0.85 (*E. coli* SEC470), 0.71 (*D. melanogaster*), and 0.58 (*H. sapiens*).
See also Figure S3 and Table S3.

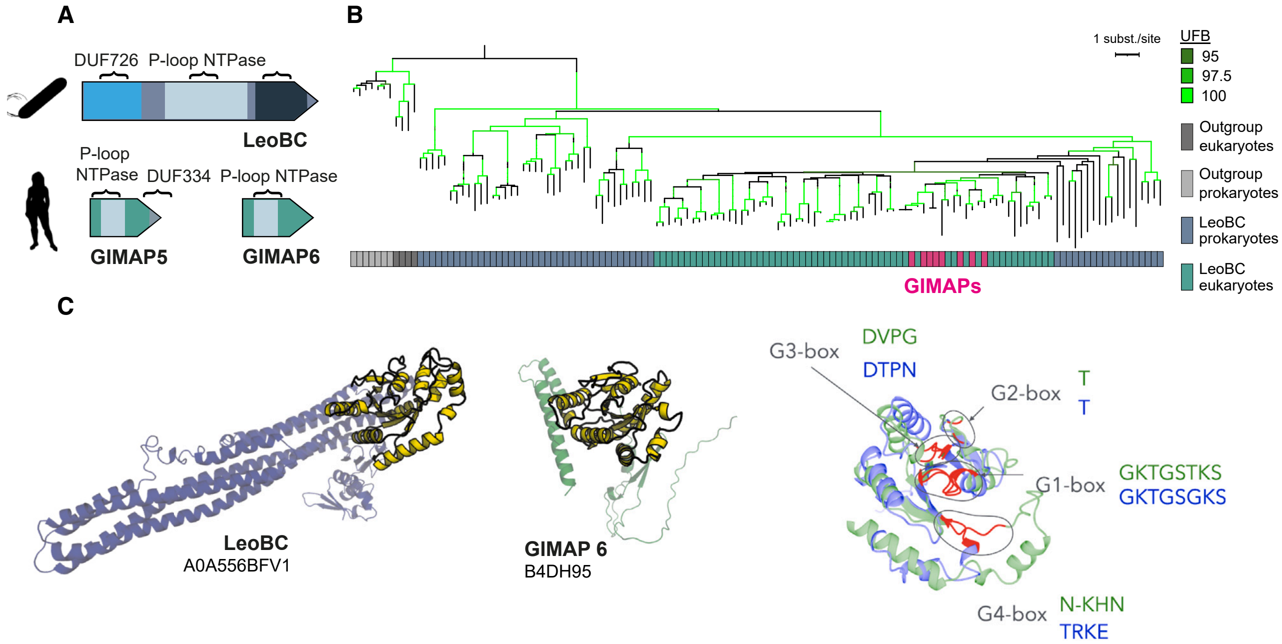

Figure 4. Human GIMAPs are likely homologs of antiphage Eleos

(A) Depiction of the LeoBC protein of antiphage Eleos, as well as its human homologs GIMAP5 and 6. The P-loop NTPase domain is conserved in GIMAPs.

(B) Phylogeny of bacterial LeoBC and their eukaryotic proteins with sequence similarity, including human GIMAPs (pink). Ultra-fast bootstraps >95 are highlighted in green. Full tree in Figure S4A.

(C) Structural comparison of prokaryotic LeoBC with human GIMAP6. TM score = 0.4. Structures were predicted using AlphaFold2 (Table S3). Conservation of active residues between LeoBC (UniProt: A0A556BFV1, green) and GIMAP6 (UniProt: B4DH95, blue) along the G1 to G4 box motifs (highlighted in red) is indicated.

See also Figure S4 and Table S3.

in which we replaced the GTPase domain of GIMAP6 by the G1 to G5 boxes of LeoA (Figure S5F). The GIMAP-Eleos chimera shows a 4-fold decrease in the percentage of HSV-infected cells, highlighting the functional conservation between Eleos and GIMAPs (Figure 5E).

Collectively, these data show that GIMAPs, a group of ISGs that likely originate from antiphage Eleos, thwart HSV-1 infection.

Proteins similar to Lamassu are encoded in microsynteny and confer antiviral activity against herpesvirus

We next focused on likely homologs of the Lamassu antiphage system. Lamassu is composed of LmuB, which encodes for a coil-coiled domain-binding DNA, as well as LmuA, a predicted effector that can be a trypsin protease (Figure 6A). The system may also include a third protein, LmuC, with no identified domains. In humans, chymotrypsin-C (CTRC) is detected with our method as a hit of LmuA (Figures 6A, 6B, and S6A). The human *CTRC* gene is located next to Forkhead-associated phosphopeptide binding domain 1 (*FHAD1*), whose protein product is also an LmuB hit (Figure 6A). *FHAD1* and *CTRC* genes are separated by EF-hand domain-containing protein D2 (*EFHD2*), which protein product performs pleiotropic functions, including regulation of immune activation in myeloid cells.[50] Akin to a prototypical bacterial genetic organization, human genes coding for proteins similar to LmuA and B are genomically co-localized, displaying microsynteny (Figure 6A). While it is not expected to observe microsynteny conserved across large evolutionary distances, eukaryotic homologs of viperin are located next to *CMPK2* kinase gene and function together to provide antiviral activity.[22] Such genetic organization has also been described in multiple bacterial and other eukaryotic viperin loci, suggesting that the viperin/kinase functional genetic linkage may have been conserved in prokaryotes and eukaryotes.[9] When exploring the conservation of the Lamassu microsynteny across eukaryotes, we could document 345 examples of loci encoding proteins similar to both LmuA and LmuB (Figures 6C and 6D). Among chordates, which include humans, 35% of genomes display LmuA/LmuB microsynteny, compared with a single example of MkoA/MkoB microsynteny. The existence of *FHAD1* and *CTRC* co-localization, together with the documented interaction of EFHD2 with intact capsids of HSV-1, suggests that FHAD1, EFHD2, and CTRC may function together in antiviral immunity.[46]

Transcriptomic profiles of immune cells indicate that FHAD1, EFHD2, and CTRC are expressed in naive B cells and dendritic cells in human and mice, a type of myeloid cell responsible for immune surveillance and priming of the adaptive immune response (Figures 6E and S6B). This hints towards a possible cooperation between these three proteins. We evaluated the potential antiviral role of FHAD1, EFHD2, and CTRC using the gain-of-function experimental setup described in Figure 5A. Expression of EFHD2 and CTRC was detected upon plasmid transfection in 293T cells, while FHAD1 was undetectable (Figure S6C). We thus focused on EFHD2 and CTRC, relying on the fact that

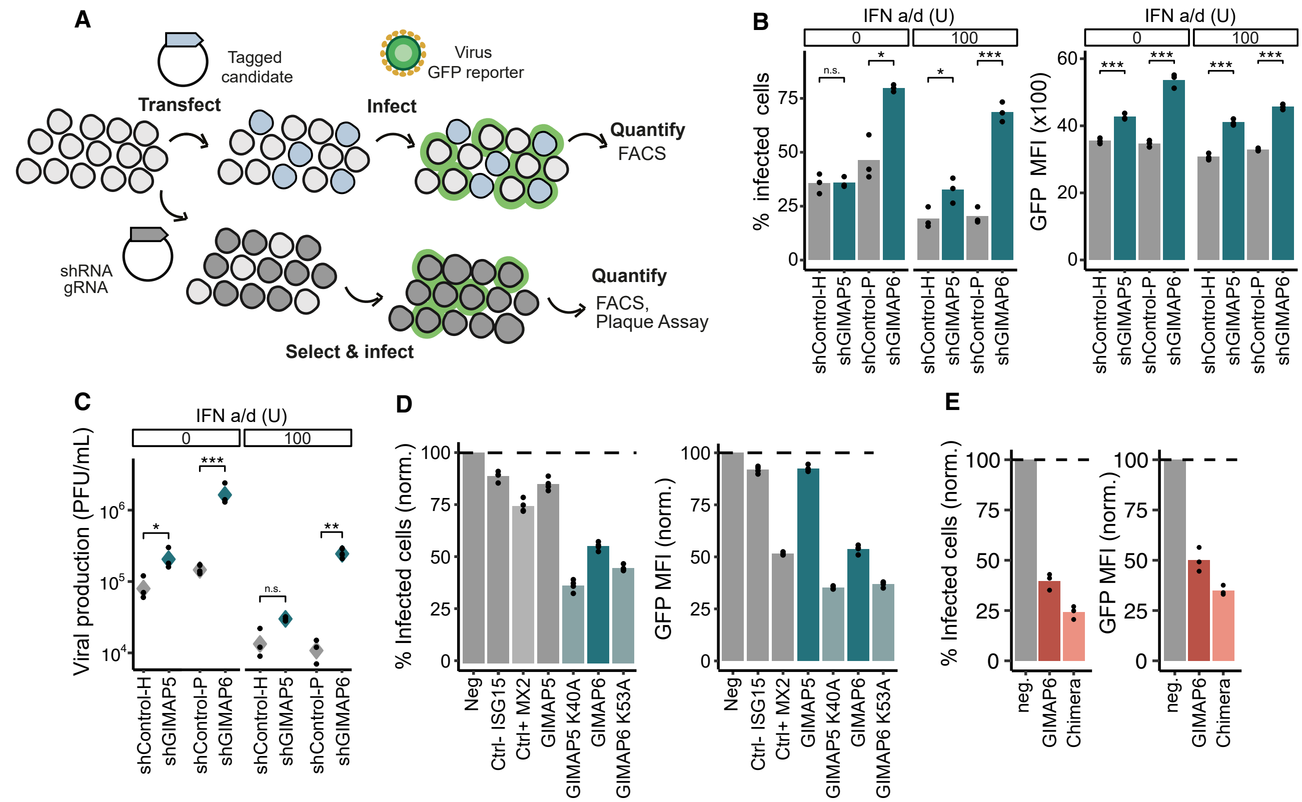

Figure 5. Human GIMAP5 and 6 are antiviral against HSV-1

(A) In experiments (B) and (C), GIMAP5 and 6 expression is knocked down in human HeLa cells by stable shRNA transfection. In (D) and (E), candidate proteins are expressed in human 293T cells via plasmid transfection. Cells are infected with HSV-1 coding for GFP. Candidates' expression and viral infection are monitored on a per cell basis by flow cytometry, and viral particle production is measured by plaque assay.

(B) Knockdown of GIMAP5 or 6 expression in human HeLa cells by shRNA transduction. Cells are infected with HSV-1, with or without pre-treatment with 100 U of IFNa/d for 8 h. Percentage of infected cells and GFP intensity are quantified by flow cytometry at 48 h post-infection. *n* = 3 biological replicates, Student's t test, unpaired.

(C) Viral particle production in the supernatant of HeLa cells measured by plaque assay at 48 h post-infection. *n* = 3 biological replicates.

(D) 293T cells transfected with plasmids encoding for control proteins involved in immunity (ISG15, negative control, and MX2, positive control), antiviral candidates (GIMAP5 and GIMAP6), or associated mutants (GIMAP5-K40A and GIMAP6-K53A), and infected with HSV-1. Percentage of infected cells and GFP intensity are quantified by flow cytometry at 48 h post-infection. Values for cells expressing the indicated protein (x axis) are normalized on values for cells with undetectable protein expression (neg, paired data from a single population of transfected cells). *n* = 4 biological replicates. Left, paired t test comparing cells expressing no protein (Neg) and cells expressing ISG15 ($p = 0.006$), MX2 ($p = 0.009$), GIMAP5 ($p = 0.002$), and GIMAP6 ($p = 0.0007$). t test between normalized value of wild-type and associated mutant, GIMAP5 versus GIMAP5 K40A ($p = 4e-16$), and GIMAP6 versus GIMAP6 K53A ($p = 1e-5$). Right, GFP mean fluorescence intensity (MFI) of cells expressing the indicated protein (x axis) is normalized by the GFP MFI of infected cells with undetectable protein expression. Paired t test between populations with and without ISG15 ($p = 0.004$), MX2 ($p = 3e-5$), GIMAP5 ($p = 0.002$), or GIMAP6 ($p = 5e-5$). t test between normalized GFP MFI in wild type and mutant, GIMAP5 versus GIMAP5-K40A ($p < 2e-16$), and GIMAP6 versus GIMAP6-K53A ($p = 1e-12$).

(E) 293T cells transfected with plasmids encoding for GIMAP6 or a chimera of GIMAP6 and Eleos (GTP-binding domain switch) and infected with HSV-1. Percentage of infected cells and GFP intensity are quantified by flow cytometry at 48 h post-infection. Values for cells expressing the indicated protein (x axis) are normalized on values for cells with undetectable protein expression (neg, paired data from a single population of transfected cells). *n* = 3 biological replicates. Left, paired t test comparing cells expressing no protein (Neg) and cells expressing GIMAP6 ($p = 0.001$), chimera ($p = 5e-5$). Right, paired t test comparing the GFP MFI between cells expressing no protein and cells expressing GIMAP6 ($p = 0.007$), chimera ($p = 6e-4$). n.s., non-significant, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

See also Figure S5.

CTRC likely bears an effector function through its protease domain, as demonstrated for bacterial LmuA.[3] Because CTRC is a protease secreted by pancreatic exocrine cells, it is not expected to accumulate inside transfected cells and should display limited intracellular immunostaining detected by flow cytometry.[51] Indeed, cells expressing only CTRC but not EFHD2 represent 2.4% of total transfected cells (Figures S6D and S6E). By contrast, 30% of transfected cells express EFHD2 alone, and 29% express both CTRC and EFHD2, suggesting that EFHD2 presence allows the intracellular retention of CTRC (Figure S6E). We then infected cells to assess the proteins' antiviral function. Cells expressing CTRC + EFHD2 display a 50% reduction in HSV-1 infection and GFP MFI compared with non-transfected cells and cells expressing only EFHD2 (Figure 6F). No antiviral effect was observed against vesicular stomatitis virus (Figure S6F).

To interrogate the role of endogenous FHAD1 and CTRC in thwarting HSV-1 infection, we first utilized myeloid cells derived from human blood, in which gene expression is knockdown by lentivirus-mediated shRNA integration. Infection with HSV-1-GFP is monitored by live video-microscopy (Figures S6G and

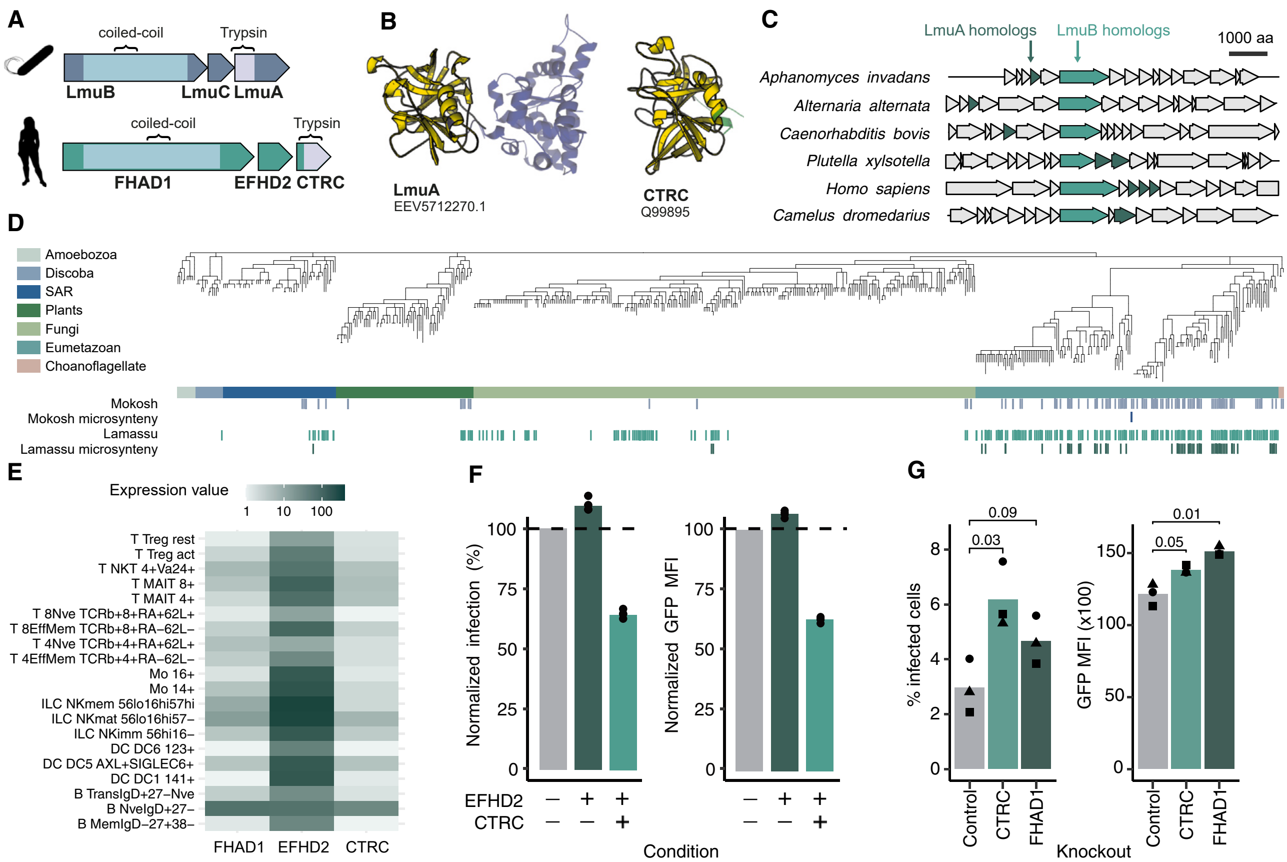

Figure 6. Human proteins likely homologs of antiphage Lamassu are antiviral

(A) FHAD1 and CTRC are human proteins likely homologs of LmuB and A, of antiphage Lamassu, detected by our pipeline, which show conservation of the coiled-coil and trypsin domains, respectively. EFHD2 is intercalated between FHAD1 and CTRC.

(B) Structural comparison of LmuA from a Lamassu of *E. coli* with human CTRC, TM score = 0.69. (AlphaFold2, Table S3).[35]

(C) Genomic context of different eukaryotic homologs of Lamassu.

(D) NCBI taxonomic tree of eukaryotic genomes. Each leaf is a species. Branch length corresponds to the number of taxonomical level associated with a species. Detection of eukaryotic proteins with sequence similarity to Mokosh (MkoA and MkoB) and Lamassu (LmuA and LmuB) is represented in blue and green, respectively. The second colored line indicates which systems are present in microsynteny.

(E) Expression of FHAD1, EFHD2, and CTRC in human immune cells (RNA-seq data from the Human Cell Atlas—immune system, population-averaged gene expression values).

(F) 293T cells co-transfected with plasmids encoding for CTRC and EFHD2 and infected with HSV-1 (multiplicity of infection [MOI] = 0.5). Infection is monitored by flow cytometry at 48 h. Percentage of infected cells expressing the indicated protein (x axis) is normalized on the percentage of infected cells in cells with undetectable expression of the indicated protein (paired data from a single population of transfected cells). $n = 4$ biological replicates, paired t test for CTRC-EFHD2 ($p = 7.8e{-5}$). Corresponding GFP MFI plots are shown in Figure S6D.

(G) THP-1 cell knockout for CTRC or FHAD1 infected with HSV-1 (MOI = 5). Infection is monitored by flow cytometry at 24 h. Percentage of infected cells (left) or GFP mean fluorescence intensity (MFI, right). $n = 3$ biological replicates, paired t test for CTRC versus control ($p = 0.03$, left and $p = 0.05$, right) and FHAD1 versus control ($p = 0.09$, left and $p = 0.01$, right).

See also Figure S6 and Table S3.

S6H). Lentivirus treatment translates into variable levels of increase in HSV-1 infection (Figure S6I). Addition of B18R (an IFN inhibitor) in the culture media does not prevent increased infection, suggesting that the antiviral effect is independent of type I IFNs. Note that the observed antiviral effect was donor-dependent, which may relate to the differences in genetic backgrounds and/or life history.

We thus decided to explore the antiviral role of FHAD1 and CTRC in an orthogonal model by CRISPR-Cas-mediated knockout (KO) of the genes in THP-1, a human macrophage cell line. FHAD1 and CTRC THP-1 KO cells display 1.5- and 2-fold increase in percentage of infected cells compared with wild type (Figure 6G). Altogether, these results document the conservation of microsynteny of likely homologs of Lamassu across eukaryotes and demonstrate that human proteins likely homolog of the antiphage Lamassu system protect against HSV-1.

DISCUSSION

In this work, we leverage the use of conservation of immune proteins across domains of life to discover immune genes in humans. These data, as well as previous work, show that proteins similar to antiviral defense mechanisms in bacteria are involved at multiple levels of eukaryotic immune pathways, including pathogen detection (cGAS/STING), signal transduction (TIR domains), activity against transposons (PLD6/MOV10L1), and antiviral effectors (GIMAPs, FHAD1/EFHD2/CTRC, viperin).[8,9,11,15,26,52] Bacterial antiphage systems may have therefore provided building blocks for the evolution of immunity in eukaryotes. Importantly, the fact that certain hits uncovered by our analysis display antiviral activity does not imply that all hits will have a role in immunity. Instead, a significant proportion may have been co-opted for unrelated cellular functions or may present a dual role within and outside of immunity. Evolutionary and mechanistic studies will help delineate each protein's contribution to eukaryotic immune pathways. We did not perform an exhaustive analysis of proteins similar to antiphage systems but rather chose to focus on specific systems. It is thus likely that additional proteins linked to antiphage systems will be identified.

Our data document the existence of proteins that are likely homologs of full antiphage systems, composed of multiple proteins. First, there is a conservation of protein-protein interaction without microsynteny, as exemplified with eukaryotic homologs of Mokosh. The interaction between MkoA and MkoB may have been conserved across billions of years of evolution to give rise to PLD6-MOV10L1 of the animal piRNA pathway. Second, we document conservation of microsynteny for likely homologs of the Lamassu system in specific eukaryotic clades, including chordates. Such non-canonical eukaryotic genetic organization suggests the existence of selective pressures to maintain Lamassu homologs under tight co-transcriptional control. It is tempting to speculate that eukaryotic Lamassu could be akin to the toxin-antitoxin systems of bacteria, in which the toxin's activity needs to be inhibited by the antitoxin to prevent cell demise. Mechanistic studies will help understanding the maintenance of microsyntenies within eukaryotic genomes. From an evolutionary perspective, the system was either directly inherited from bacteria as a whole (present in the last eukaryotic common ancestor or acquired through horizontal gene transfer after the emergence of eukaryotes), or these proteins were selected to participate in the same pathway during evolution. In the case of FHAD1 and CTRC, this would also imply the selection for microsynteny.

Our work shows that human proteins likely originating from Eleos (GIMAPs) and Lamassu (FHAD1/EFHD2/CTRC) can shield cells from herpesviruses, which are part of the *Duplodnaviridae*, the realm of viruses that includes bacteriophages.[53] Connections between phages and herpesviruses were documented by a recent work describing the antiphage system Avs, which can be triggered *in vitro* by human herpesvirus-8 proteins.[12] Whether GIMAPs and FHAD1/CTRC share antiviral molecular mechanisms with their cognate antiphage system, targeting conserved structures of the *Duplodnaviridae* family, common to bacterial phages and human herpesviruses, will be the topic of further studies.

Beyond the proof of concept in humans, our work provides a list of likely homologs of antiphage systems in thousands eukaryotic organisms. We propose here that comparative genomics of immune proteins across domains of life, in addition to unraveling striking prokaryote-eukaryote conservation, could uncover new mechanisms of defense across eukaryotes, including plants or protists. Such a method could be a starting point in the study of immunology in non-model organisms, for which the experimental toolbox is limited.

To account for the ever-growing number of immune proteins conserved across the tree of life, we propose the concept of ancestral immunity.[54] Ancestral immune modules are conserved proteins or protein domains performing an immune function in prokaryotes and eukaryotes. The refinement of our method of ancestral immunity exploration, as well as the expansion of knowledge on antiphage systems, will provide a more comprehensive understanding of prokaryotes' contribution to eukaryotic immunity.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.chom.2024.08.002.

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AUTHOR CONTRIBUTIONS

All the authors designed experiments and analyzed data. J.C., E.M., M.H., A.H., M.R., and F.T. performed data curation and formal analysis. V.H.T., M.H., D.B., and A.H. performed experiments. J.C., E.M., M.H., G.O., E.Z.P., and A.B. wrote the manuscript, with contributions from M.R. and L.Q.-M. P.B., A.B., E.Z.P., and L.Q.-M. acquired funding.

DECLARATION OF INTERESTS

A patent application related to this work has been filled (EP22306832.1).

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-Flag	BIOLEGEND	RRID: AB_2563148
Anti-HA	BIOLEGEND	RRID: AB_2566616
Bacterial and virus strains		
HSV-1 VP26-GFP, KOS strain	Desai et al., https://doi.org/10.1128/JVI.72.9.7563-7568.1998	N/A
VSV-GFP	Silvin et al., https://doi.org/10.1126/sciimmunol.aai8071	N/A
Biological samples		
Human blood from healthy donors	Etablissement Français du Sang	N/A
Chemicals, peptides, and recombinant proteins		
Universal Type I IFN Protein	R&D Systems	11200-1
Nigericin	Invivogen	tlrl-nig
Staurosporine	R&D Systems	1285/100U
Critical commercial assays		
Cell permeabilization kit	Thermofisher scientific	GAS004
Vialbility dye	Thermofisher scientific	65-0863-18
Experimental models: Cell lines		
HEK293T cells	ATCC	ATCC-CRL-3216
HeLa cells	Kind gift from Benaroch Lab	N/A
THP-1 cells	ATCC	ATCC-TIB-202
Recombinant DNA		
pcDNA 3.1 (+) HA CTRC	Genscript	N/A
pcDNA 3.1 (+) FLAG EFHD2	Genscript	N/A
pcDNA 3.1 (+) FLAG GIMAP 5	Genscript	N/A
pcDNA 3.1 (+) HA GIMAP 6	Genscript	N/A
pcDNA 3.1 (+) FLAG MXB	Genscript	N/A
pcDNA 3.1 (+) V5 FHAD1	Genscript	N/A
pcDNA 3.1 (+) HA ISG15	Genscript	N/A
Software and algorithms		
Software for flow cytometry	FLOWJO	FLOWJO 10.9.0
Other		
GCCCAACCTATCCGCCCGAG LentiCRISPRv2 Puromycin CTRC gRNA targeting exon 2	Genscript	N/A
CCTCCACATAAATCTCCGCA LentiCRISPRv2 Puromycin FHAD1 gRNA targeting exon 5	Genscript	N/A
CCTGGGTTAGAGCTACCGCA LentiCRISPRv2 Puromycin Scrambled gRNA to generate Control cells	Genscript	N/A
GCGCGCTCACCAGATCATAAA pLKO.1 Puromycin FHAD1 shRNA	Sigma	N/A
GACTTTGATTGCTAGCAACTT pLKO.1 Puromycin CTRC shRNA	Sigma	N/A
CAACAAGATGAAGAGCACCAA pLKO.1 Puromycin Scramble shRNA shC002	Sigma	N/A

(*Continued on next page*)

Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
GGACCCTGGAGCACTTCTAAT pLV[shRNA]-Hygro-U6 Hygromycin GIMAP5 shRNA	VectorBuilder	N/A
CCTAAGGTTAAGTCGCCCTCG pLV[shRNA]-Hygro-U6 Hygromycin Scramble shRNA	VectorBuilder	N/A
CCACCAAGGTAGACTTAGATT pLV[shRNA]-Puro-U6 Puromycin GIMAP6 shRNA	VectorBuilder	N/A
CCTAAGGTTAAGTCGCCCTCG pLV[shRNA]-Puro-U6 Puromycin Scramble shRNA	VectorBuilder	N/A

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to the lead contact, Aude Bernheim (aude.bernheim@pasteur.fr).

Materials availability

All unique reagents generated during this study will be available upon request.

Data and code availability

- RNAseq used in this work originates from the ImmGen project (https://www.immgen.org/). Data including all alignments, profiles and trees are available at https://github.com/mdmparis/cury_mordret_et_al_supplementary_data.
- This study does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available form the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Viral stocks and plaque assay

Herpes Simplex virus 1 (HSV-1, KOS strain) was built by fusing VP26 fused to GFP.[55] Viral stocks were amplified on 293T cells and titrated by plaque assay. Briefly, 200,000 Vero cells were seeded in 24-well plates and infected with 10-fold dilutions of sample for 1h at 37°C. Cells were then overlaid with DMEM containing 2% FCS, 100 U/ml P/S and 0.8% agarose. After 4 days, cells were fixed with 10% formalin and visualised by crystal violet staining.

Lentivirus

Lentivirus used to infect human-blood derived monocytes and THP-1 cells were produced by transfecting 293LTV cells (plated in 75cm2 flasks) with 8 ug of DNA and 200uL of PEI MAX (Polysciences). Per flask, transfected DNA is constituted of 1,2 ug pMD2.G (Addgene #12259), 2,8 ug psPAX2 (Addgene #12260), 0,15 ug Vpr-and-Vpx and 4 ug of either LentiCRISPRv2 or pLV [shRNA] / pLKO.1-puro. LentiCRISPRv2 plasmids encoding gRNA against CTRC and FHAD1 were purchased from Genscript. PLKO.1-Puro plasmids coding for a shRNA targeting human FHAD1 transcripts (TRCN0000253814), human CTRC transcripts (TRCN0000047025) or a control shRNA (shC002) were purchased from Merck.[56] Media was changed 16h after transfection and supernatant was harvested at 2 days post transfection, aliquoted and frozen. Lentivirus used to knockdown GIMAP5 (VB240115-1191aau), GIMAP6 (VB240115-1180trj), and a scrambled control (VB010000-9466ycn) were purchased from VectorBuilder.

Viral infections

HEK 293T cells were infected with HSV-GFP at a multiplicity of infection (MOI) of 0.5 one day after plasmid transfection (see plasmid transfections) and collected 48 hours post-infection for flow cytometry analysis. For VSV experiments, HEK 293T cells were infected with VSV-GFP at an MOI of 0.01 two days after plasmid transfection and collected 5 hours post-infection for flow cytometry analysis. HeLa cells were infected with HSV-GFP at an MOI of 0.5 8 hours after treatment with 100 units of universal type I interferon (PBL Assay Science). Cells were washed with PBS one-hour post-infection and fresh media was added. Both supernatant and cells were collected 48 hours post-infection for titration (see *Viral Stocks and Plaque Assay*) and flow cytometry analysis. THP-1 cells were infected with HSV-GFP after being differentiated with 50 ng/mL phorbol 12-myristate-13-acetate (PMA) for 24 hours, washed with PBS, and left to rest for 72 hours in fresh media. Cells were collected 24h post-infection for flow cytometry analysis.

Generation of human blood-derived myeloid cells and infection

We obtained plasmapheresis residues of healthy adult donors from the Etablissement Français du Sang (Paris, France), in line with ethical guidelines of the Institut National de la Santé et de la Recherche Médicale (INSERM, France) and with French Public Health laws (art. L 1121-1-1, art. L1121-1-2). Institutional Review Board and obtention of written consent are not required when performing human non-interventional studies.

Peripheral blood mononuclear cells (PBMCs) were separated from whole blood using Ficoll-Paque (GE Healthcare), and monocytes were enriched using CD14+ magnetic beads (Miltenyi Biotec). Isolated cells were transduced with lentivirus-containing media mixed with 8 ug/ml protamine (Merck) and differentiated into macrophages for 10 days by culture in RPMI (Gibco) with 5% FCS, 5% human serum AB (Sigma Aldrich), 1% P/S and 25 ng/ml M-CSF (Miltenyi Biotec). Three days after transduction, cells were selected with 3 ug/ml puromycin. Differentiated cultures were infected with HSV-GFP at MOI 3, and infection was followed by live video-microscopy using an Incucyte system (Sartorius). Images were analysed using the Incucyte analysis software. After a top-hat segmentation step, fluorescence intensity of each object was measured in each condition and integrated with the surface of GFP positive cells in each well. Experiment performed in duplicates for each donor.

METHOD DETAILS

Database of eukaryotic and prokaryotic genomes

Eukaryotic genomes were downloaded from genbank (https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/) in January 2022. Isoforms were removed from the protein fasta files, when possible, by keeping the longest one. Pseudogenes were removed as well, when annotated as such. CDS were renamed to take into account the relative position to easily assess whether two proteins are co-localized in the genome. In total, 4,616 eukaryotic genomes representing 2,407 species were screened. 22,920 prokaryotic complete genomes were downloaded from Refseq (https://ftp.ncbi.nih.gov/genomes/refseq/bacteria/) in July 2022.

Identification of proteins with sequence similarity to antiphage systems in eukaryotic genomes

DefenseFinder v1.0.9 with models v1.2.2 was run with default parameters on the custom database of 4,616 eukaryotic genomes, generating 14,701 candidate systems composed of a total of 17,932 associated genes. The output hits of DefenseFinder was examined to remove potential bacterial genomic contaminations. Each candidate gene was searched against the UniProt90 (https://www.uniprot.org) database using Diamond (v2.0.9 -b 10 -c 1 --max-target-seqs 5 --outfmt 6), retrieving the 5 best hits per query.[57] We used the uniprot id-mapping API to fetch the common ncbi taxon id of each of the UniProt90 families. This step fetched a taxon id for 78% of the hits. For the remaining 22% of the hits whose protein's id did not match any UniProt90 family, we fetched their taxonomy by mapping the id to UniProtKB (12%) or to UniParc (10%). Each candidate protein was deemed a contamination if at least one of the 5 best hits on the UniProt90/UniProtKB/UniParc database came from a prokaryote, and its associated systems were subsequently removed from the pipeline. This approach led us to remove a total of 746 (4.2%) genes and 484 systems (3.3%) from our list, leaving us with 14,217 eukaryotic system candidates composed of 17,186 genes (Table S4).

Homology search of Mokosh, Eleos, Lamassu

We looked for sequence similarity in eukaryotic genomes starting from the following primary HMM profiles: LmuA_effector_Protease, LmuB_SMC_Hydrolase_protease, Mokosh_TypeI_MkoA, Mokosh_TypeI_MkoB, Dynamins_LeoA (now called Eleos__LeoA), Dynamins_LeoBC (now called Eleos__LeoBC) (available at https://github.com/mdmparis/defence-finder-models). For each profile, the pipeline for homology search was similar. First, we searched the primary profile against our custom database of eukaryotic and prokaryotic proteins using hmmsearch (version 3.3.2), at an e-value threshold of 1e-3.[58] Prokaryotic hits were then marked as "involved in immunity" if they were found by DefenseFinder, and as "isolated" otherwise. We selected a subset of these sequences (sequence datasets available at https://github.com/mdmparis/cury_mordret_et_al_supplementary_data), containing approximately 200 paired and 300 isolated prokaryotic hits, to perform downstream phylogenetic analyses. For both the paired and isolated groups, we picked half of the proteins among the top ranking hits to the reference hmm profile, and the other half by sampling uniformly along the list.

The subset of sequences was then filtered to remove duplicated proteins (using usearch v11.0.667 with the option -cluster_fast and -id 1) and aligned with mafft (version 7.505 with the option --auto).[59,60] The resulting alignment was trimmed with ClipKit (version 1.3.0) using the kpic-gappy mode to keep informative and constant sites and removes sites with more than 90% of gaps.[61] Phylogenetic trees were computed using IQTree (version 2.2.0.3) using model VT+F+I+G4 and 1000 ultrafast bootstraps, and visualised on the iTOL webserver (https://itol.embl.de).[62] The trees were further annotated to display whether hits were involved in immunity or not, and whether the sequences were eukaryotic or prokaryotic. As documented in the sequence datasets, (https://github.com/mdmparis/cury_mordret_et_al_supplementary_data), clades were manually selected to include eukaryotic proteins distant from prokaryotic proteins with non-immune function (Figures 1A, 1B, and S1B) . Eukaryotic members of selected clades were aligned using mafft (with parameter --auto). We performed a manual curation of the alignments by removing sequences that did not align well with the rest, recomputed the alignments, and manually truncated the N and C termini. These alignments were used to build secondary hmm profiles using hmmbuild (from hmmer version 3.3.2). All alignments, profiles, weblogos and trees are available at https://github.com/mdmparis/cury_mordret_et_al_supplementary_data.

Each of these secondary profiles was then used to search against the eukaryotic protein database using hmmsearch. We visualised the distribution of scores, and manually determined a score threshold to keep or discard these secondary hits (Figure "Score distribution used to select hits based on eukaryotic HMM profiles" deposited at https://github.com/mdmparis/cury_mordret_et_al_supplementary_data). We identify outgroups for MkoA, MkoB and LeoBC using Foldseek and selected proteins that had the best match with a good annotation score on Uniprot (>4/5) and that are present in both eukaryotes and prokaryotes. The outgroup of MkoA, MkoB and LeoBC are respectively SMUBP-2, ClsB and ERA G-protein. We performed phmmer starting from the human homologs and selected hits intersecting with the one obtained with the new eukaryotic profile. Along with the outgroups and the selected eukaryotic proteins, we selected prokaryotic proteins identified by DefenseFinder and removed redundancy at 50% using mmseqs2 (easy-linclust --min-seq-id 0.5). For MkoB, we conserved hits with profile coverage of more than 40% to be more selective and hit by the PLDc_2 profile, and kept the corresponding MkoA proteins. For the three proteins, alignments were built using either mafft or Muscle version 5.1 (with --super5 parameter), trimmed using ClipKit (with model gappy at 95%), and the trees were built with IQ-tree (-m LG+G4 -B 1000).[61,63] The trees shown in Figures 3B and 4B were manually pruned to lighten the visual representation. Complete trees of MkoA, MkoB and LeoBC are shown in Figures S3 and S4.

In our method, phylogenetic trees are only used to obtain a reduced number of hits employed to build HMM profiles. Such profiles are then utilized in the next iteration of homology search. Opposite to Jackhmmer, which considers all the hits above a defined threshold for the next iteration, we select here a subset of the hits guided by phylogeny. Unfortunately, phylogenetic trees of HMM hit spanning the tree of life are difficult to build and thus not always well resolved. We decided to focus instead on building a new HMM profile with hits that appear either "close" to true bacterial hit, or at least "far" from false bacterial hits. It is possible we could have chosen more eukaryotic hits from the tree, but this would have likely led to too many hits to be experimentally tested. In some cases, other clades could have probably been selected instead of our clade of choice.

Structural comparisons

Structures were either downloaded from the alphafold.ebi.ac.uk database, or computed using the ColabFold notebook.[35,64,65] We created custom databases of structures and searched them using the easy-search command of Foldseek version 3.915ef7d (Table S3, https://github.com/mdmparis/cury_mordret_et_al_supplementary_data.) with default parameters.[28] Structures were imported, realigned based on the Foldseek alignment, and visualised using PyMol version 2.5.4 (Schrödinger, LLC).[66]

Cofold of structures

Structures of bacterial Mokosh and their predicted eukaryotic homologs were co-folded using AlphaFold 2.3.1 in the multimer, reduced_db, default settings. The relaxed structures with the highest ptm+iptm score were selected and visualised with ChimeraX 1.6.1.[67]

Cell culture

Human embryonic kidney 293T cells (293T cells) and Vero cells were grown in Dulbecco's modified Eagle's medium (DMEM, Gibco) including 10% fetal calf serum (FCS) and 100U/ml penicillin/streptomycin (P/S, Gibco). Prior to transfection, 293T cells were plated in medium without antibiotics.

Human evolutionary analysis

To quantify selective constraint genome-wide, we downloaded hg19 base-wise Rejected Substitutions GERP scores[68] from http://mendel.stanford.edu/SidowLab/downloads/gerp/, excluded sites with GerpRS equal to 0 (missing alignment data), and computed for each gene, the mean GerpRS across coding exons, as defined in Ensembl75 (hg19). Gene-wise evidence of adaptive evolution was quantified based on the branch-site Random effects likelihood model (BS-REL) which detects diversifying selection by quantifying the percentage of codons where the rate of non-synonymous substitutions is greater than the rate of synonymous substitutions (dN/dS>1) in each branch of a phylogeny.[30] Specifically, we retrieved precomputed proportion of selected codons detected by BS-REL for 9,861 genes across the mammalian phylogeny[69] (Table S5) and considered for each gene the average of these proportions across all branches of the phylogeny. We then focused on the set of 9,268 genes where both GerpRS and BS-REL scores where available and contrasted these two metrics between genes identified by our method (n = 48) or jackhmmer (n = 464), defence response genes defined from gene ontology (either antiviral - GO:0051607, antibacterial - GO:0042742, or antifungal - GO:0050832, n = 176 in total), or all annotated genes (n = 9,268). Pairwise comparisons were performed with a two-sided Wilcoxon rank sum test. We evaluated the evidence for stronger positive selection (BS-REL) on our pipeline candidates across various branches of the mammalian tree (primates, glires, cetartiodactyla & zooamata, Figure 2C).

Lentivirus infection and selection

HeLa cells were transduced with pLV[shRNA] lentiviral particles at a multiplicity of infection (MOI) of 3 in the presence of 1 μg/mL polybrene, followed by the addition of fresh media supplemented with 1 μg/mL puromycin (for shGIMAP6 and shControl-Puro) or 500 μg/mL hygromycin (for shGIMAP5 and shControl-Hygro) 48 hours post-transduction. The cells were grown in selective media until the non-transduced cells died, which took 3 to 4 days, and then plated in non-selective media before undergoing interferon treatment and viral infection.

THP-1 cells were transduced with LentiCRISPRv2 lentiviral particles at a MOI of 5 in the presence of 1 μg/mL polybrene, and fresh media supplemented with 3 μg/mL puromycin was added 48 hours post-transduction. After 5 days, the cells were serially diluted into 96-well plates, and wells containing a single cell were identified and allowed to grow for 3 weeks. The resulting clones were expanded and verified by nanopore sequencing of the PCR amplicons around the target locus (see Supplementary Data in https://github.com/mdmparis/cury_mordret_et_al_supplementary_data).

Plasmid transfections

Sequences coding for human proteins of interest were synthesised through Genscript and integrated in a pcDNA3.1(+) backbone. GIMAP5 and 6 were tagged with a flag tag and a HA tag at the N-terminus, respectively, whilst EFHD2 was equipped with a flag tag (N-terminus), CTRC with a HA tag (C-terminus) and FHAD1 with a V5 tag (C-terminus). 100 ng of each plasmid was transfected in 293T cells using Lipofectamine 2000 (Thermo Fisher Scientific) following the manufacturer's instruction.

Flow cytometry

Cells were collected in a V-shape bottom 96-well plate (Corning) and resuspended in 50 ul PBS supplemented with DAPI. FIX & PERM kit (Thermo Fisher Scientific, GAS004) was used according to the manufacturer's instructions. Proteins of interest were visualised using an anti-Flag-PE (BioLegend, 637310), an anti-HA-Alexa647 (BioLegend, 682404) and an anti-Flag-PE-Cy7 (Thermo Fisher Scientific, 25-6796-42) at a 1:300 dilution. Analyses were performed on a FACSVerse Cell Analyser (BD Biosciences) and results were analysed with FlowJo (TreeStar). Between 30,000 and 150,000 events were recorded and analysed from each well.

Data visualisation

Trees were drawn with ITOL (https://itol.embl.de).[70] Schematics of eukaryotic organisms were taken from https://beta.phylopic.org/. Graphs were plotted using ggplot2, matplotlib (https://matplotlib.org) and seaborn (http://seaborn.pydata.org/index.html). Web logos of alignment were built using weblogo.[71]

QUANTIFICATION AND STATISTICAL ANALYSIS

Quantifications and statistics were performed as described in the methods section and figure legends. Legends contain number of biological replicates, name of the test, whether the data is paired or not, and any associated P-values. Statistical analysis was performed with GraphPad Prism v10 or with R version 4.2 and Rstudio v2023.06.2. For data frame manipulations, we used dplyr v1.0.10 along with data.table v1.12.8. No method was used to determine whether the data met assumptions of the statistical approach. Data were collected from at least n = 3 independent experiments and are shown as bar plots (mean) with individual points (biological replicates), or violin plots showing 1st and 3rd quartiles, split by the median (line), with whiskers extending a 1.5-fold interquartile range beyond the box.